

Karthus project 2026

The sensitivity of grounding-line retreat to ice-shelf damage

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Introduction

Mass loss of the Antarctic ice sheet is essentially localized in marine basins where the ice sheet is in contact with the ocean and where warm ocean water penetrate underneath the floating ice shelves. This project gauges the sensitivity of marine ice sheets to reduction in ice shelf buttressing by ice damage (brittle failure) and its effect on mass loss and grounding line migration. In order to investigate this sensitivity, we will employ the Kori-ULB ice sheet model. The group will focus on an idealized marine ice sheet geometry, for which previously damage experiments were carried out (Sun et al., 2017).

The Kori-ULB ice sheet model

The Kori-ULB ice sheet model is written in Matlab and can be downloaded from <https://github.com/FrankPat/Kori-ULB>. Either download all files and subroutines, make the path available and things should run. To test, just type KoriModel in the prompt and the model should run a square ice sheet (with graphical output).

You can further use one of the scripts from <https://github.com/FrankPat/Karthus> to do some further testing of simple geometries. The PDF file with the manual can also be downloaded and gives a comprehensive overview on how to use the ice sheet model.

Experimental setup

We will use a synthetic setup of a marine terminating glacier with a geometry that mimics the geometry of Pine Island Glacier, Antarctica. The geometry has been defined in Gudmundsson et al. (2012) and is used for the MISMP+ experiments described in Cornford et al. (2020).

- Initialize the model with a steady-state grounding line positioned on the upward sloping bed (see script for details). Use the hybrid model in order to make a comparison with the real-world experiments possible.
- Perform sensitivity experiments by including ice damage.
- Sensitivity of sliding law (regularized Coulomb versus Weertman)
- Sensitivity to parameters in the damage parameterization.

References

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